

# Formal Verification: Liquid Staking and Yield Protocol

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## Abstract

We formalize the liquid staking model where staked assets earn yield while remaining tradeable. The key properties are monotone yield accrual, no impermanent loss on single-sided staking, and compositional compounding.

## 1 Introduction

The liquid staking protocol enables “money at rest” to earn yield. Staked assets receive validator rewards while remaining tradeable via liquid derivative tokens (xLUX). Auto-compounding reinvests rewards into the staking position.

## 2 Definitions

**Definition 1** (Position). *Staking position: principal, accrued rewards, compounded rewards, liquid derivatives, duration.*

## 3 Theorems and Axioms

**Theorem 2** (yield\_monotone). *Accrued rewards only increase:  $(accrueBlock\ p).accrued \geq p.accrued$ .*

**Theorem 3** (derivatives\_equal\_principal). *Liquid derivatives are minted 1:1 with principal.*

**Theorem 4** (compound\_resets\_accrued). *After compounding, accrued resets to 0.*

**Theorem 5** (compound\_increases\_base). *Compounding increases the effective principal.*

**Theorem 6** (no\_impermanent\_loss). *Single-sided staking has no impermanent loss: principal is unchanged by accrual.*

**Theorem 7** (value\_monotone). *Total value (principal + compounded + accrued) is monotonically non-decreasing.*

## 4 Lean Source

*/-*

*Liquid Staking and Yield Protocol*

*Staked assets earn yield while remaining liquid.*

*Money at rest is put to work - implicit staking of idle liquidity in the DEX pools.*

*Maps to:*

- *lux/node: validator staking (P-Chain)*
- *lux/exchange: LP staking rewards*
- *zoo/contracts: liquid staking derivatives*

*Properties:*

- *Yield accrual: staked assets earn validator rewards*
- *Liquidity: staked assets remain tradeable (liquid derivatives)*
- *No impermanent loss on single-sided staking*
- *Compounding: rewards auto-compound into staking position*

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*-/*

```
import Mathlib.Data.Nat.Defs
import Mathlib.Tactic
```

```
namespace DeFi.LiquidStaking
```

```
-- Staking position -/
```

```
structure Position where
```

```
  principal : Nat          -- original staked amount
  accrued   : Nat          -- accumulated rewards
  compounded : Nat         -- rewards that have been re-staked
  liquidDerivatives : Nat  -- tradeable staking receipt tokens
  stakeDuration : Nat      -- blocks staked
```

```
-- Yield rate: annual percentage in basis points (e.g., 500 = 5%) -/
```

```
def annualYieldBps : Nat := 500 -- 5% APY baseline
```

```
-- Per-block yield (simplified: annual / ~2.6M blocks per year) -/
```

```
def blockYield (principal : Nat) : Nat :=
  principal * annualYieldBps / 26000000
```

```
-- Accrue one block of yield -/
```

```
def accrueBlock (p : Position) : Position :=
  let yield := blockYield (p.principal + p.compounded)
  { p with accrued := p.accrued + yield
    , stakeDuration := p.stakeDuration + 1 }
```

```
-- Compound: move accrued rewards into principal -/
```

```
def compound (p : Position) : Position :=
  { p with compounded := p.compounded + p.accrued
    , accrued := 0 }
```

```
-- YIELD ACCRUAL: Accrued rewards only increase -/
```

```
theorem yield_monotone (p : Position) :
  (accrueBlock p).accrued > p.accrued := by
  simp [accrueBlock]; omega
```

```
-- LIQUIDITY: Liquid derivatives equal principal (1:1 mint) -/
```

```

def mintDerivatives (p : Position) : Position :=
  { p with liquidDerivatives := p.principal }

theorem derivatives_equal_principal (p : Position) :
  (mintDerivatives p).liquidDerivatives = p.principal := by
  simp [mintDerivatives]

/-- COMPOUNDING: After compound, accrued resets to 0 -/
theorem compound_resets_accrued (p : Position) :
  (compound p).accrued = 0 := by
  simp [compound]

/-- COMPOUNDING: After compound, compounded increases -/
theorem compound_increases_base (p : Position) :
  (compound p).compounded p.compounded := by
  simp [compound]; omega

/-- NO IMPERMANENT LOSS: Single-sided staking has no IL.
    Unlike LP positions, staked assets don't change ratio. -/
theorem no_impermanent_loss (p : Position) :
  (accrueBlock p).principal = p.principal := by
  simp [accrueBlock]

/-- DURATION MONOTONE: Stake duration only increases -/
theorem duration_monotone (p : Position) :
  (accrueBlock p).stakeDuration > p.stakeDuration := by
  simp [accrueBlock]; omega

/-- TOTAL VALUE: principal + compounded + accrued -/
def totalValue (p : Position) : Nat :=
  p.principal + p.compounded + p.accrued

/-- TOTAL VALUE MONOTONE: Accruing never decreases total -/
theorem value_monotone (p : Position) :
  totalValue (accrueBlock p) totalValue p := by
  simp [totalValue, accrueBlock]; omega

/-- MONEY AT REST: Idle liquidity earns yield.
    DEX pool deposits implicitly stake via the liquid staking layer. -/
theorem idle_earns (p : Position) (h : p.principal > 0) :
  totalValue (accrueBlock p) totalValue p := value_monotone p

end DeFi.LiquidStaking

```

## 5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/DeFi/LiquidStaking.lean>

White papers: <https://papers.lux.network>

Related white papers:

- lux-liquid-staking.tex
- lux-restaking.tex